Important Functions in Computer Science

Ceiling and Floor Functions, Step Graphs, Recursive Functions, and Sequence Recurrences

1 Ceiling and Floor Functions

In computer science, we frequently bridge the gap between continuous real-valued quantities ($x \in \mathbb{R}$) and discrete integer resources ($n \in \mathbb{Z}$). For example, storing a file of x bytes into disk blocks of 4096 bytes requires an integer number of blocks, and splitting an array of n elements in binary search requires an integer midpoint index. Two foundational rounding functions handle this conversion: the **ceiling function** and the **floor function**.

DEFINITION **Ceiling Function**

The **ceiling function** assigns to the real number $x \in \mathbb{R}$ the **smallest integer that is greater than or equal to** x.

- **Notation**:

$$f(x) = \lceil x \rceil$$

- **Characterization**: For any $x \in \mathbb{R}$ and $n \in \mathbb{Z}$,

$$\lceil x \rceil = n \iff n - 1 < x \leq n$$

DEFINITION **Floor Function**

The **floor function** assigns to the real number $x \in \mathbb{R}$ the **largest integer that is less than or equal to** x. (The floor function is also known as the **greatest integer function**.)

- **Notation**:

$$f(x) = \lfloor x \rfloor$$

- **Characterization**: For any $x \in \mathbb{R}$ and $n \in \mathbb{Z}$,

$$\lfloor x \rfloor = n \iff n \leq x < n + 1$$

EXAMPLE & WALKTHROUGH **Ceiling and Floor Function Evaluations**

- **Ceiling Function Examples**:
  - $\lceil 6.76 \rceil = 7$ because 7 is the smallest integer ≥ 6.76 ($6 < 6.76 \leq 7$).
  - $\lceil 5.1 \rceil = 6$ because 6 is the smallest integer ≥ 5.1 ($5 < 5.1 \leq 6$).
- **Floor Function Examples**:
  - $\lfloor 6.76 \rfloor = 6$ because 6 is the largest integer ≤ 6.76 ($6 \leq 6.76 < 7$).
  - $\lfloor 5.1 \rfloor = 5$ because 5 is the largest integer ≤ 5.1 ($5 \leq 5.1 < 6$).

1.1 Step-Function Graphs and Algebraic Properties

Both $y = \lfloor x \rfloor$ and $y = \lceil x \rceil$ are **piecewise-constant step functions** mapping $\mathbb{R}$ to $\mathbb{Z}$. On each unit interval between consecutive integers n and $n + 1$:

- The **floor function** $y = \lfloor x \rfloor$ stays constant at height n on the **left-closed, right-open** interval $[n, n + 1)$, jumping up by 1 at the right endpoint $x = n + 1$.
- The **ceiling function** $y = \lceil x \rceil$ stays constant at height $n + 1$ on the **left-open, right-closed** interval $(n, n + 1]$, jumping up by 1 immediately to the right of $x = n$.

Comparison: Floor Function vs. Ceiling Function

Floor Function: y = floor(x)

- **Direction**: Rounds **down** toward $-\infty$ on the number line.
- **Constant Steps**: $\lfloor x \rfloor = n$ for $x \in [n, n+1)$ (closed point on the left at (n, n), open point on the right at $(n+1, n)$).
- **Bounds**: $x - 1 < \lfloor x \rfloor \leq x$.

Ceiling Function: y = ceil(x)

- **Direction**: Rounds **up** toward $+\infty$ on the number line.
- **Constant Steps**: $\lceil x \rceil = n$ for $x \in (n-1, n]$ (open point on the left at $(n-1, n)$, closed point on the right at (n, n)).
- **Bounds**: $x \leq \lceil x \rceil < x + 1$.

Behavior at Exact Integers (x in Z)

If and only if $x \in \mathbb{Z}$, both functions equal x:

$$\lfloor x \rfloor = \lceil x \rceil = x \iff x \in \mathbb{Z}$$

When $x \notin \mathbb{Z}$, we always have $\lceil x \rceil - \lfloor x \rfloor = 1$.

Useful Algebraic Identities (for x in R, k in Z)

- **Integer Shift**: $\lfloor x + k \rfloor = \lfloor x \rfloor + k$ and $\lceil x + k \rceil = \lceil x \rceil + k$.
- **Negation Duality**:

$$\lfloor -x \rfloor = -\lceil x \rceil \quad \text{and} \quad \lceil -x \rceil = -\lfloor x \rfloor$$

Interval of x	Floor $\lfloor x \rfloor$	Ceiling $\lceil x \rceil$	Step-Graph Endpoint Behavior
$-3 \leq x < -2$	-3	-3 (at $x = -3$) or -2	Floor closed at $(-3, -3)$, open at $(-2, -3)$
$-2 \leq x < -1$	-2	-2 (at $x = -2$) or -1	Floor closed at $(-2, -2)$, open at $(-1, -2)$
$-1 \leq x < 0$	-1	-1 (at $x = -1$) or 0	Floor closed at $(-1, -1)$, open at $(0, -1)$
$0 \leq x < 1$	0	0 (at $x = 0$) or 1	Floor closed at $(0, 0)$, open at $(1, 0)$
$1 \leq x < 2$	1	1 (at $x = 1$) or 2	Floor closed at $(1, 1)$, open at $(2, 1)$
$2 \leq x < 3$	2	2 (at $x = 2$) or 3	Floor closed at $(2, 2)$, open at $(3, 2)$

PITFALL / WARNING **Negative Numbers in Floor and Ceiling Functions**

A common mistake is confusing `floor(x)` with simply dropping the fractional part (truncation toward zero).

- For **positive** numbers, $\lfloor 2.99 \rfloor = 2$ does match dropping .99.
- For **negative** numbers, $\lfloor -0.1 \rfloor = -1$ (not 0) because $-1 \leq -0.1 < 0$, and -1 lies to the **left** of -0.1 on the number line. Conversely, $\lceil -0.1 \rceil = 0$ (not -1) because 0 lies to the **right** of -0.1.

When in doubt, picture the number line: **floor moves left**, **ceiling moves right**.

EXAMPLE & WALKTHROUGH **Practice Problems: Evaluating Floor and Ceiling Expressions**

Set 1 — Find these values:

- **a)** $\lfloor \mathbf{1.1} \rfloor$: Since $1 \leq 1.1 < 2$, we have $\lfloor \mathbf{1.1} \rfloor = \mathbf{1}$.
- **b)** $\lceil \mathbf{1.1} \rceil$: Since $1 < 1.1 \leq 2$, we have $\lceil \mathbf{1.1} \rceil = \mathbf{2}$.
- **c)** $\lfloor \mathbf{-0.1} \rfloor$: Since $-1 \leq -0.1 < 0$, rounding down (left) gives $\lfloor \mathbf{-0.1} \rfloor = \mathbf{-1}$.
- **d)** $\lceil \mathbf{-0.1} \rceil$: Since $-1 < -0.1 \leq 0$, rounding up (right) gives $\lceil \mathbf{-0.1} \rceil = \mathbf{0}$.
- **e)** $\lceil \mathbf{2.99} \rceil$: Since $2 < 2.99 \leq 3$, we have $\lceil \mathbf{2.99} \rceil = \mathbf{3}$.
- **f)** $\lceil \mathbf{-2.99} \rceil$: Since $-3 < -2.99 \leq -2$, rounding up (right) toward zero gives $\lceil \mathbf{-2.99} \rceil = \mathbf{-2}$.

- **g)** $\left\lfloor \frac{1}{2} + \left\lceil \frac{1}{2} \right\rceil \right\rfloor$: Evaluate from the inside out: first $\left\lceil \frac{1}{2} \right\rceil = \lceil 0.5 \rceil = 1$. Substituting into the outer floor gives:

$$\left\lfloor \frac{1}{2} + \left\lceil \frac{1}{2} \right\rceil \right\rfloor = \left\lfloor \frac{1}{2} + 1 \right\rfloor = \lfloor 1.5 \rfloor = \mathbf{1}$$

- **h)** $\left\lceil \left\lfloor \frac{1}{2} \right\rfloor + \left\lceil \frac{1}{2} \right\rceil + \frac{1}{2} \right\rceil$: First evaluate the inner terms: $\left\lfloor \frac{1}{2} \right\rfloor = 0$ and $\left\lceil \frac{1}{2} \right\rceil = 1$. Substituting into the outer ceiling gives:

$$\left\lceil 0 + 1 + \frac{1}{2} \right\rceil = \lceil 1.5 \rceil = \mathbf{2}$$

Set 2 — Find these values:

- **a)** $\left\lceil \frac{3}{4} \right\rceil$: Since $0 < 0.75 \leq 1$, we have $\left\lceil \frac{3}{4} \right\rceil = \mathbf{1}$.
- **b)** $\left\lfloor \frac{7}{8} \right\rfloor$: Since $0 \leq 0.875 < 1$, we have $\left\lfloor \frac{7}{8} \right\rfloor = \mathbf{0}$.
- **c)** $\left\lceil -\frac{3}{4} \right\rceil$: Since $-1 < -0.75 \leq 0$, we have $\left\lceil -\frac{3}{4} \right\rceil = \mathbf{0}$.
- **d)** $\left\lfloor -\frac{7}{8} \right\rfloor$: Since $-1 \leq -0.875 < 0$, we have $\left\lfloor -\frac{7}{8} \right\rfloor = \mathbf{-1}$.
- **e)** $\lceil \mathbf{3} \rceil$: Since $3 \in \mathbb{Z}$ is already an integer, $\lceil \mathbf{3} \rceil = \mathbf{3}$.
- **f)** $\lfloor \mathbf{-1} \rfloor$: Since $-1 \in \mathbb{Z}$ is already an integer, $\lfloor \mathbf{-1} \rfloor = \mathbf{-1}$.
- **g)** $\left\lfloor \frac{1}{2} + \left\lceil \frac{3}{2} \right\rceil \right\rfloor$: First evaluate the inner ceiling: $\left\lceil \frac{3}{2} \right\rceil = \lceil 1.5 \rceil = 2$. Then:

$$\left\lfloor \frac{1}{2} + \left\lceil \frac{3}{2} \right\rceil \right\rfloor = \lfloor 0.5 + 2 \rfloor = \lfloor 2.5 \rfloor = \mathbf{2}$$

- **h)** $\left\lfloor \frac{1}{2} \cdot \left\lfloor \frac{5}{2} \right\rfloor \right\rfloor$: First evaluate the inner floor: $\left\lfloor \frac{5}{2} \right\rfloor = \lfloor 2.5 \rfloor = 2$. Multiplying by $\frac{1}{2}$ inside the outer floor gives:

$$\left\lfloor \frac{1}{2} \cdot \left\lfloor \frac{5}{2} \right\rfloor \right\rfloor = \left\lfloor \frac{1}{2} \cdot 2 \right\rfloor = \lfloor 1 \rfloor = \mathbf{1}$$

2 Recursive Functions

Sometimes it is difficult or awkward to define a sequence or function directly with a closed-form formula, whereas it is natural to describe how each value builds on earlier values. This approach is called **recursion**.

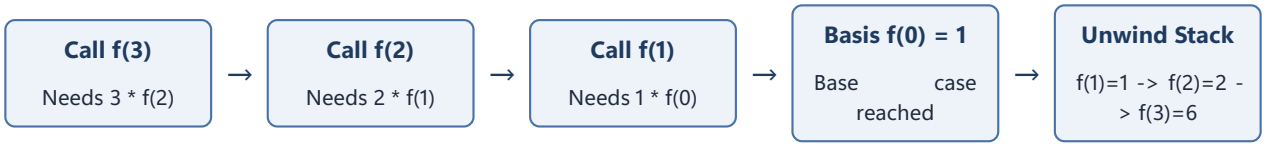
DEFINITION **Recursive Function**

A function f (typically with domain $\mathbb{N} = \{0, 1, 2, \ldots\}$ or $\mathbb{Z}^+ = \{1, 2, 3, \ldots\}$) is called a **recursive function** if its definition refers to itself. Every well-defined recursive function consists of two essential parts:

1. **Basis Step**: Specifies the initial value(s) of the function directly, **without** referring to the function itself (for example, $f(0)$ or $f(1)$).
2. **Recursive (or Inductive) Step**: Defines the value of the function at the current input n in terms of its own values at smaller inputs (such as $f(n-1), f(n-2), \ldots$).

INTUITION & DEEP DIVE **Visualizing Recursion: Self-Similarity and Call Stack Unwinding**

Recursion is like a picture-within-a-picture (the **Droste effect**), where a scene contains a smaller copy of the exact same scene, which in turn contains an even smaller copy. In programming and mathematics, however, an endless chain of self-references without a stopping condition causes infinite recursion (a stack overflow). The **Basis Step** acts as the innermost frame that holds a concrete value instead of another recursive call, allowing the evaluation to stop and unwind back to the original call.



Evaluating f(3) for a recursive function with a base case at f(0).

2.1 Classic Recursive Functions

2.1.1 The Factorial Function

Let $f(n) = n!$ denote the factorial function on $n \in \mathbb{N}$. Because $n! = n \cdot (n-1) \cdot (n-2) \cdots 2 \cdot 1 = n \cdot (n-1)!$ for $n \geq 1$, we can write $f(n)$ recursively as:

$$f(n) = \begin{cases} 1 & \text{, for } n = 0 \quad \text{(Basis)} \\ n \cdot f(n-1) & \text{, for } n > 0 \quad \text{(Recurrence)} \end{cases}$$

EXAMPLE & WALKTHROUGH **Step-by-Step Evaluation of the Factorial Function**

Starting from the basis at $n = 0$ and stepping forward for $n = 0, 1, 2, 3, \ldots$:

- $n = 0 \implies f(0) = 1 = 0!$ (by the basis step)
- $n = 1 \implies f(1) = 1 \cdot f(0) = 1 \cdot 1 = 1 = 1!$
- $n = 2 \implies f(2) = 2 \cdot f(1) = 2 \cdot 1 = 2 = 2!$
- $n = 3 \implies f(3) = 3 \cdot f(2) = 3 \cdot 2 \cdot 1 = 6 = 3!$
- $n = 4 \implies f(4) = 4 \cdot f(3) = 4 \cdot 6 = 24 = 4!$, and so on.

2.1.2 Chebyshev Polynomials

A recursive definition can depend on **two** (or more) previous terms. When a recurrence depends on both $n-1$ and $n-2$ (a second-order recurrence), we must provide **two basis cases** ($n = 0$ and $n = 1$) so that the recurrence has enough starting information at $n = 2$.

The **Chebyshev polynomials of the first kind**, denoted $T(n, x)$ (used widely in numerical analysis and approximation theory because they satisfy $T(n, \cos\theta) = \cos(n\theta)$), are defined recursively by:

$$T(n, x) = \begin{cases} 1 & , \quad n = 0 \\ x & , \quad n = 1 \\ 2xT(n-1, x) - T(n-2, x) & , \quad n > 1 \end{cases}$$

EXAMPLE & WALKTHROUGH **Evaluating Chebyshev Polynomials**

This function has two basis cases: $n = 0$ and $n = 1$. Computing the first few polynomials:

- $n = 0 \implies T(0, x) = 1$
- $n = 1 \implies T(1, x) = x$
- $n = 2 \implies$ Substituting $n = 2$ into the recurrence:

$$\begin{aligned} T(2, x) &= 2xT(1, x) - T(0, x) \\ &= 2x(x) - 1 \\ &= 2x^2 - 1 \end{aligned}$$

- $n = 3 \implies$ Substituting $n = 3$ into the recurrence:

$$T(3, x) = 2xT(2, x) - T(1, x) = 2x(2x^2 - 1) - x = 4x^3 - 3x$$

2.1.3 The Fibonacci Sequence

The **Fibonacci function** is another classic second-order recurrence relation with two basis values ($n = 0$ and $n = 1$):

$$f(n) = \begin{cases} 0 & , \ n = 0 \\ 1 & , \ n = 1 \\ f(n-1) + f(n-2) & , \ n > 1 \end{cases}$$

EXAMPLE & WALKTHROUGH **Evaluating Fibonacci Numbers for n = 0, 1, 2, 3, 4, 5**

Problem: Find the values of the Fibonacci function $f(n)$ for $n = 0, 1, 2, 3, 4, 5$.

Solution: Starting from the basis values $f(0) = 0$ and $f(1) = 1$, each subsequent term is the sum of the two preceding terms:

- $\boldsymbol{n = 0}$: $f(0) = \mathbf{0}$ **(Basis 1)**
- $\boldsymbol{n = 1}$: $f(1) = \mathbf{1}$ **(Basis 2)**
- $\boldsymbol{n = 2}$: $f(2) = f(1) + f(0) = 1 + 0 = \mathbf{1}$
- $\boldsymbol{n = 3}$: $f(3) = f(2) + f(1) = 1 + 1 = \mathbf{2}$
- $\boldsymbol{n = 4}$: $f(4) = f(3) + f(2) = 2 + 1 = \mathbf{3}$
- $\boldsymbol{n = 5}$: $f(5) = f(4) + f(3) = 3 + 2 = \mathbf{5}$

3 Expressing Closed-Form Functions Recursively

To convert an explicit formula $f(n)$ or sequence a_n into a recursive definition, we follow two steps:

1. **Formulate the Recurrence**: Express the current term $f(n)$ (or a_n) using the previous term $f(n-1)$ (or a_{n-1}), either by expanding and grouping repeated operations or by algebraic substitution.
2. **Specify the Basis**: Evaluate the formula at the starting index of the domain (typically $n = 0$ or $n = 1$).

3.1 Recursive Definition of Powers $f(n) = a^n$

Suppose we want to express $f(n) = a^n$ (for $a \neq 0$ and $n \in \mathbb{N}$) recursively:

1. **Express $f(n)$ in terms of $f(n-1)$**:

$$f(n) = a^n = \underbrace{a \cdot a \cdot a \cdots a}_{n \text{ factors}} = a \cdot \underbrace{(a \cdot a \cdots a)}_{n-1 \text{ factors}} = a \cdot a^{n-1} = a \cdot f(n-1)$$

2. **Determine the basis**: At $n = 0$, we have $f(0) = a^0 = 1$.
3. **Complete Recursive Definition**:

$$f(n) = \begin{cases} 1 & , \ n = 0 \\ a \cdot f(n-1) & , \ n > 0 \end{cases}$$

3.2 Recursive Definition of $a_n = 6n$ via Two Methods

Suppose we want to express the linear sequence $a_n = 6n$, for $n = 1, 2, 3, \ldots$, recursively. We can derive the recurrence in two ways.

EXAMPLE & WALKTHROUGH **Deriving the Recurrence for a_n = 6n**

Method 1 — Expansion and Grouping:

- **Recurrence**: Write multiplication by n as repeated addition of 6 (n times), then group the last $n-1$ terms together:

$$\begin{aligned} a_n = 6n &= \underbrace{6 + 6 + \ldots + 6}_{n \text{ times}} \\ &= 6 + \underbrace{(6 + \ldots + 6)}_{n-1 \text{ times}} \\ &= 6 + 6(n-1) \\ &= 6 + a_{n-1} \end{aligned}$$

- **Basis**: Because n starts at $n = 1$, the basis is:

$$a_1 = 6(1) = 6$$

Method 2 — Algebraic Substitution from the Closed Form:

- Write down the closed-form formula at index n and at index $n-1$:

$$a_n = 6n \quad \text{and} \quad a_{n-1} = 6(n-1) = 6n - 6$$

- Substitute $a_{n-1} = 6n - 6$ into the expression for a_n:

$$a_n = 6n = (6n - 6) + 6 = a_{n-1} + 6$$

- **Complete Recursive Definition**:

$$a_n = \begin{cases} 6 & , \ n = 1 \\ 6 + a_{n-1} & , \ n > 1 \end{cases}$$

INTUITION & DEEP DIVE **Using Finite Differences to Formulate Any Recurrence**

Method 2 generalizes to **any** sequence a_n:

- **Additive (Difference) Form**: Compute the difference between consecutive terms, $\Delta a_n = a_n - a_{n-1}$. Once simplified, rewrite it as:

$$a_n = a_{n-1} + (a_n - a_{n-1})$$

  For example, if $a_n = n^2$, then $a_n - a_{n-1} = n^2 - (n-1)^2 = 2n - 1$, which immediately gives $a_n = a_{n-1} + 2n - 1$.
- **Multiplicative (Ratio) Form**: For geometric sequences where $\frac{a_n}{a_{n-1}} = r$, rewrite as $a_n = r \cdot a_{n-1}$.

4 Worked Practice Exercises

Below are complete, step-by-step solutions to the practice problems on evaluating and formulating recursive functions.

4.1 First-Order Recurrence Evaluations

EXAMPLE & WALKTHROUGH **Evaluating First-Order Recurrences (Set 1)**

Problem: Find $f(1), f(2), f(3)$, and $f(4)$ if $f(n)$ is defined recursively by $\boldsymbol{f(0) = 1}$ and for $n = 0, 1, 2, \ldots$:

- **a) $\boldsymbol{f(n+1) = f(n) + 2}$ (Arithmetic progression with common difference $+2$)**:
  - $f(1) = f(0) + 2 = 1 + 2 = \mathbf{3}$
  - $f(2) = f(1) + 2 = 3 + 2 = \mathbf{5}$
  - $f(3) = f(2) + 2 = 5 + 2 = \mathbf{7}$
  - $f(4) = f(3) + 2 = 7 + 2 = \mathbf{9}$
- **b) $\boldsymbol{f(n+1) = 3f(n)}$ (Geometric progression with common ratio 3)**:
  - $f(1) = 3f(0) = 3(1) = \mathbf{3}$
  - $f(2) = 3f(1) = 3(3) = \mathbf{9}$
  - $f(3) = 3f(2) = 3(9) = \mathbf{27}$
  - $f(4) = 3f(3) = 3(27) = \mathbf{81}$
- **c) $\boldsymbol{f(n+1) = 2^{f(n)}}$ (Iterated exponentiation)**:
  - $f(1) = 2^{f(0)} = 2^1 = \mathbf{2}$
  - $f(2) = 2^{f(1)} = 2^2 = \mathbf{4}$

- $f(3) = 2^{f(2)} = 2^4 = \mathbf{16}$
- $f(4) = 2^{f(3)} = 2^{16} = \mathbf{65{,}536}$

- **d) $f(n+1) = f(n)^2 + f(n) + 1$ (Quadratic recurrence)**:
  - $f(1) = f(0)^2 + f(0) + 1 = 1^2 + 1 + 1 = \mathbf{3}$
  - $f(2) = f(1)^2 + f(1) + 1 = 3^2 + 3 + 1 = 9 + 3 + 1 = \mathbf{13}$
  - $f(3) = f(2)^2 + f(2) + 1 = 13^2 + 13 + 1 = 169 + 13 + 1 = \mathbf{183}$
  - $f(4) = f(3)^2 + f(3) + 1 = 183^2 + 183 + 1 = 33,489 + 183 + 1 = \mathbf{33{,}673}$

EXAMPLE & WALKTHROUGH **Evaluating First-Order Recurrences (Set 2)**

Problem: Find $f(1), f(2), f(3), f(4)$, and $f(5)$ if $f(n)$ is defined recursively by $\boldsymbol{f(0) = 3}$ and for $n = 0, 1, 2, \ldots$:

- **a) $f(n+1) = -2f(n)$**:
  - $f(1) = -2f(0) = -2(3) = \mathbf{-6}$
  - $f(2) = -2f(1) = -2(-6) = \mathbf{12}$
  - $f(3) = -2f(2) = -2(12) = \mathbf{-24}$
  - $f(4) = -2f(3) = -2(-24) = \mathbf{48}$
  - $f(5) = -2f(4) = -2(48) = \mathbf{-96}$
- **b) $f(n+1) = 3f(n) + 7$**:
  - $f(1) = 3(3) + 7 = 9 + 7 = \mathbf{16}$
  - $f(2) = 3(16) + 7 = 48 + 7 = \mathbf{55}$
  - $f(3) = 3(55) + 7 = 165 + 7 = \mathbf{172}$
  - $f(4) = 3(172) + 7 = 516 + 7 = \mathbf{523}$
  - $f(5) = 3(523) + 7 = 1569 + 7 = \mathbf{1576}$
- **c) $f(n+1) = f(n)^2 - 2f(n) - 2$**:
  - $f(1) = 3^2 - 2(3) - 2 = 9 - 6 - 2 = \mathbf{1}$
  - $f(2) = 1^2 - 2(1) - 2 = 1 - 2 - 2 = \mathbf{-3}$
  - $f(3) = (-3)^2 - 2(-3) - 2 = 9 + 6 - 2 = \mathbf{13}$
  - $f(4) = 13^2 - 2(13) - 2 = 169 - 26 - 2 = \mathbf{141}$
  - $f(5) = 141^2 - 2(141) - 2 = 19,881 - 282 - 2 = \mathbf{19{,}597}$
- **d) $f(n+1) = 3^{\frac{f(n)}{3}}$**:
  - $f(1) = 3^{\frac{f(0)}{3}} = 3^{\frac{3}{3}} = 3^1 = \mathbf{3}$
  - $f(2) = 3^{\frac{f(1)}{3}} = 3^{\frac{3}{3}} = 3^1 = \mathbf{3}$
  - $f(3) = 3^{\frac{f(2)}{3}} = 3^{\frac{3}{3}} = \mathbf{3}$
  - $f(4) = 3^{\frac{f(3)}{3}} = 3^{\frac{3}{3}} = \mathbf{3}$
  - $f(5) = 3^{\frac{f(4)}{3}} = 3^{\frac{3}{3}} = \mathbf{3}$

(Notice that $f(0) = 3$ is a fixed point of the map $x \mapsto 3^{\frac{x}{3}}$, so $f(n) = 3$ for all $n \geq 0$.)

4.2 Second-Order Recurrence Evaluations

EXAMPLE & WALKTHROUGH **Evaluating Second-Order Recurrences (Set 3)**

Problem: Find $f(2), f(3), f(4)$, and $f(5)$ if f is defined recursively by $\boldsymbol{f(0) = -1, f(1) = 2}$, and for $n = 1, 2, \ldots$:

- **a) $f(n+1) = f(n) + 3f(n-1)$**:
  - $n = 1 \Longrightarrow f(2) = f(1) + 3f(0) = 2 + 3(-1) = 2 - 3 = \mathbf{-1}$
  - $n = 2 \Longrightarrow f(3) = f(2) + 3f(1) = -1 + 3(2) = -1 + 6 = \mathbf{5}$

- $n = 3 \Rightarrow f(4) = f(3) + 3f(2) = 5 + 3(-1) = 5 - 3 = \mathbf{2}$
- $n = 4 \Rightarrow f(5) = f(4) + 3f(3) = 2 + 3(5) = 2 + 15 = \mathbf{17}$

- **b) $f(n+1) = f(n)^2 f(n-1)$:**
  - $n = 1 \Rightarrow f(2) = f(1)^2 f(0) = (2)^2(-1) = \mathbf{-4}$
  - $n = 2 \Rightarrow f(3) = f(2)^2 f(1) = (-4)^2(2) = 16 \cdot 2 = \mathbf{32}$
  - $n = 3 \Rightarrow f(4) = f(3)^2 f(2) = (32)^2(-4) = 1024 \cdot (-4) = \mathbf{-4096}$
  - $n = 4 \Rightarrow f(5) = f(4)^2 f(3) = (-4096)^2(32) = 16,777,216 \cdot 32 = \mathbf{536{,}870{,}912}$ (or 2^{29})
- **c) $f(n+1) = 3f(n)^2 - 4f(n-1)^2$:**
  - $n = 1 \Rightarrow f(2) = 3f(1)^2 - 4f(0)^2 = 3(2)^2 - 4(-1)^2 = 12 - 4 = \mathbf{8}$
  - $n = 2 \Rightarrow f(3) = 3f(2)^2 - 4f(1)^2 = 3(8)^2 - 4(2)^2 = 192 - 16 = \mathbf{176}$
  - $n = 3 \Rightarrow f(4) = 3f(3)^2 - 4f(2)^2 = 3(176)^2 - 4(8)^2 = 3(30,976) - 256 = 92,928 - 256 = \mathbf{92{,}672}$
  - $n = 4 \Rightarrow f(5) = 3f(4)^2 - 4f(3)^2 = 3(92,672)^2 - 4(176)^2 = 25,764,298,752 - 123,904 = \mathbf{25{,}764{,}174{,}848}$
- **d) $f(n+1) = \frac{f(n-1)}{f(n)}$:**
  - $n = 1 \Rightarrow f(2) = \frac{f(0)}{f(1)} = \frac{-1}{2} = \mathbf{-1/2}$
  - $n = 2 \Rightarrow f(3) = \frac{f(1)}{f(2)} = \frac{2}{-\frac{1}{2}} = \mathbf{-4}$
  - $n = 3 \Rightarrow f(4) = \frac{f(2)}{f(3)} = \frac{-\frac{1}{2}}{-4} = \mathbf{1/8}$
  - $n = 4 \Rightarrow f(5) = \frac{f(3)}{f(4)} = \frac{-4}{\frac{1}{8}} = \mathbf{-32}$

EXAMPLE & WALKTHROUGH **Evaluating Second-Order Recurrences (Set 4)**

Problem: Find $f(2), f(3), f(4)$, and $f(5)$ if f is defined recursively by $\boldsymbol{f(0) = f(1) = 1}$ and for $n = 1, 2, \ldots$:

- **a) $f(n+1) = f(n) - f(n-1)$:**
  - $n = 1 \Rightarrow f(2) = f(1) - f(0) = 1 - 1 = \mathbf{0}$
  - $n = 2 \Rightarrow f(3) = f(2) - f(1) = 0 - 1 = \mathbf{-1}$
  - $n = 3 \Rightarrow f(4) = f(3) - f(2) = -1 - 0 = \mathbf{-1}$
  - $n = 4 \Rightarrow f(5) = f(4) - f(3) = -1 - (-1) = \mathbf{0}$

  (This sequence repeats with period 6: $\mathbf{1, 1, 0, -1, -1, 0, 1, 1, \ldots}$)
- **b) $f(n+1) = f(n)f(n-1)$:**
  - $n = 1 \Rightarrow f(2) = f(1)f(0) = 1 \cdot 1 = \mathbf{1}$
  - $n = 2 \Rightarrow f(3) = f(2)f(1) = 1 \cdot 1 = \mathbf{1}$
  - $n = 3 \Rightarrow f(4) = f(3)f(2) = 1 \cdot 1 = \mathbf{1}$
  - $n = 4 \Rightarrow f(5) = f(4)f(3) = 1 \cdot 1 = \mathbf{1}$
- **c) $f(n+1) = f(n)^2 + f(n-1)^3$:**
  - $n = 1 \Rightarrow f(2) = f(1)^2 + f(0)^3 = 1^2 + 1^3 = \mathbf{2}$
  - $n = 2 \Rightarrow f(3) = f(2)^2 + f(1)^3 = 2^2 + 1^3 = 4 + 1 = \mathbf{5}$
  - $n = 3 \Rightarrow f(4) = f(3)^2 + f(2)^3 = 5^2 + 2^3 = 25 + 8 = \mathbf{33}$
  - $n = 4 \Rightarrow f(5) = f(4)^2 + f(3)^3 = 33^2 + 5^3 = 1089 + 125 = \mathbf{1214}$
- **d) $f(n+1) = \frac{f(n)}{f(n-1)}$:**
  - $n = 1 \Rightarrow f(2) = \frac{f(1)}{f(0)} = \frac{1}{1} = \mathbf{1}$
  - $n = 2 \Rightarrow f(3) = \frac{f(2)}{f(1)} = \frac{1}{1} = \mathbf{1}$
  - $n = 3 \Rightarrow f(4) = \frac{f(3)}{f(2)} = \frac{1}{1} = \mathbf{1}$
  - $n = 4 \Rightarrow f(5) = \frac{f(4)}{f(3)} = \frac{1}{1} = \mathbf{1}$

4.3 Formulating Recursive Sequence Definitions

EXAMPLE & WALKTHROUGH **Writing Recursive Definitions for Sequences**

Set 1: Give a recursive definition of the sequence $\{a_n\}, n = 1, 2, 3, \ldots$ if:

- **a) $a_n = 6n$**:
  - **Basis**: $a_1 = 6(1) = 6$.
  - **Recurrence**: Since $a_n - a_{n-1} = 6n - 6(n-1) = 6$, we have $a_n = a_{n-1} + 6$ for $n > 1$.

$$a_n = \begin{cases} 6 & , \ n = 1 \\ a_{n-1} + 6 & , \ n > 1 \end{cases}$$

- **b) $a_n = 2n + 1$**:
  - **Basis**: $a_1 = 2(1) + 1 = 3$.
  - **Recurrence**: Since $a_n - a_{n-1} = (2n+1) - (2(n-1)+1) = 2$, we have $a_n = a_{n-1} + 2$ for $n > 1$.

$$a_n = \begin{cases} 3 & , \ n = 1 \\ a_{n-1} + 2 & , \ n > 1 \end{cases}$$

- **c) $a_n = 10^n$**:
  - **Basis**: $a_1 = 10^1 = 10$.
  - **Recurrence**: Since $a_n = 10 \cdot 10^{n-1} = 10a_{n-1}$, we have $a_n = 10a_{n-1}$ for $n > 1$.

$$a_n = \begin{cases} 10 & , \ n = 1 \\ 10a_{n-1} & , \ n > 1 \end{cases}$$

- **d) $a_n = 5$ (Constant sequence)**:
  - **Basis**: $a_1 = 5$.
  - **Recurrence**: Every term equals the previous term, so $a_n = a_{n-1}$ for $n > 1$.

$$a_n = \begin{cases} 5 & , \ n = 1 \\ a_{n-1} & , \ n > 1 \end{cases}$$

Set 2: Give a recursive definition of the sequence $\{a_n\}, n = 1, 2, 3, \ldots$ if:

- **a) $a_n = 4n - 2$**:
  - **Basis**: $a_1 = 4(1) - 2 = 2$.
  - **Recurrence**: Computing the difference gives $a_n - a_{n-1} = (4n-2) - (4(n-1)-2) = 4$. Thus $a_n = a_{n-1} + 4$ for $n > 1$.

$$a_n = \begin{cases} 2 & , \ n = 1 \\ a_{n-1} + 4 & , \ n > 1 \end{cases}$$

- **b) $a_n = 1 + (-1)^n$**: Checking the first few terms: for odd n, $a_n = 1 - 1 = 0$; for even n, $a_n = 1 + 1 = 2$. Thus the sequence alternates $0, 2, 0, 2, 0, 2, \ldots$
  - **Method A (First-Order Complement)**: Notice that $a_n + a_{n-1} = (1 + (-1)^n) + (1 + (-1)^{n-1}) = 2 + 0 = 2$. Thus $a_n = 2 - a_{n-1}$ (or $a_n = a_{n-1} + 2(-1)^n$) with basis $a_1 = 0$:

$$a_n = \begin{cases} 0 & , \ n = 1 \\ 2 - a_{n-1} & , \ n > 1 \end{cases}$$

  - **Method B (Second-Order Periodic)**: Alternatively, using two basis cases $a_1 = 0, a_2 = 2$, the recurrence is $a_n = a_{n-2}$ for $n > 2$.
- **c) $a_n = n(n+1)$**:
  - **Basis**: $a_1 = 1(1+1) = 2$.

- **Recurrence**: Compute the first difference $a_n - a_{n-1}$:

$$a_n - a_{n-1} = n(n+1) - (n-1)n = n((n+1) - (n-1)) = 2n$$

Therefore, $a_n = a_{n-1} + 2n$ for $n > 1$:

$$a_n = \begin{cases} 2 & , \quad n = 1 \\ a_{n-1} + 2n & , \quad n > 1 \end{cases}$$

- **d) $a_n = n^2$**:
  - **Basis**: $a_1 = 1^2 = 1$.
  - **Recurrence**: Compute the first difference $a_n - a_{n-1}$:

$$a_n - a_{n-1} = n^2 - (n-1)^2 = n^2 - (n^2 - 2n + 1) = 2n - 1$$

Therefore, $a_n = a_{n-1} + 2n - 1$ for $n > 1$ (showing that the n-th square is the sum of the first n odd integers):

$$a_n = \begin{cases} 1 & , \quad n = 1 \\ a_{n-1} + 2n - 1 & , \quad n > 1 \end{cases}$$